# Muhammad Ausaf Ayaz

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Portfolio: ausafayaz.github.io | GitHub: github.com/ausafayaz

*Final-year Computer Science (Data Science) student seeking a 12-week on-site IT / Software / Data internship in Karachi, Dec 2026 – Feb 2027*

## Education

**Monash University Malaysia** — Selangor, Malaysia
*Bachelor of Computer Science (Data Science)* — *Final Year – Currently Enrolled*

- **Relevant coursework:** Data Structures & Algorithms, Database Systems, Database Modelling and Management, Data Visualisation, Programming Paradigms, Intelligent Systems (AI), Foundations of Data Analysis (ETW2001)

**Karachi Grammar School** — Karachi, Pakistan
*A Levels* — *2021 – 2023*

- A* Mathematics; A Physics; A English General Paper; B Computer Science; B Chemistry

**Saint Michael's Convent School** — Karachi, Pakistan
*O Levels* — *2019 – 2021*

- 8 A* and 1 A

## Projects

More projects: github.com/ausafayaz

**Young Malaysia Under Pressure** | *Vega-Lite, JavaScript, HTML/CSS* — Live site

- Built an interactive data story of **12** Vega-Lite charts and maps from **4** official datasets (data.gov.my), covering youth unemployment, CPI and household income across **15** states and territories
- Designed a composite "pressure index" that ranks states on **3** normalised (0–100) indicators: unemployment, CPI change and spending-to-income ratio

**APAC Superstore Profitability Dashboard** | *R, Shiny, tidyverse, plotly* — GitHub

- Built an R Shiny dashboard with **5** interactive charts and **5** filters, joining **4** years (2011–14) of retail orders with **3** World Bank indicators
- Analysed **$3.59M** in APAC sales at a **12.2%** margin; showed loss-making orders rise sharply with discount level and Furniture earns the weakest margins

**Flappy Birb** | *TypeScript, RxJS, Vite, Vitest* — Play it

- Wrote a browser game in a purely functional reactive style: every input and timer is a pure reducer folded into one immutable state stream
- Implemented physics, collisions, CSV-driven levels, power-ups and ghost replays at **60** fps, covered by **18** unit tests with CI deployment on every push

## Experience

**Reet Solutions** — Jun 2022 – Jul 2022
*IT & Digital Operations Intern* — *Karachi, Pakistan*

- Supported the planning and execution of digital and social media campaigns for company and client accounts
- Scheduled and published posts across social platforms and helped keep website content up to date
- Tracked basic engagement metrics (reach, likes, follower growth) and helped compile results for the team
- Assisted with day-to-day IT and digital operations tasks, troubleshooting minor issues and organising digital assets

## Technical Skills

**Languages**: Python, Java, R, SQL, TypeScript, JavaScript, HTML, CSS
**Data & Databases**: Relational database design, SQL queries, normalisation, data cleaning, visualisation, statistical analysis
**Software Development**: Object-oriented programming, data structures & algorithms, debugging, unit testing, functional reactive programming, front-end development
**Tools & Libraries**: Pandas, NumPy, Matplotlib, R Shiny, tidyverse, plotly, Vega-Lite, RxJS, Vite, Vitest, RStudio, Git/GitHub

## Leadership & Community

**Monash University Malaysia**: Organiser and volunteer across student-led clubs and campus events, coordinating logistics and promoting events
**Karachi Grammar School**: First Aid Society member; Peer Tutoring Society tutor
**The Last Earthlings**: Volunteer, beach-cleaning community service initiative